FLOODCAST / Evaluation matrix

55 stations | 293,057 training rows | 19,306 public test rows

Selected model: ensemble_v2 Test period: 2025-01-01 to 2025-12-31

Evaluation	Rows	RMSE (cfs)	Extreme RMSE	MAE (cfs)	R-squared
Later-date validation	58,645	47,232	53,138	16,816	0.8809
Unseen stations + later dates	12,051	72,109	115,119	27,946	0.9015
2025 public benchmark (reused)	19,306	46,543	111,777	14,003	0.9149
2025 station-median baseline	19,306	85,500	180,500	25,872	0.7129

45.6% lower test RMSE than the training station-median baseline

R-squared 0.9149 describes explained variance, not a percentage of correct predictions.

Extreme RMSE uses the actual top 10% discharge days within each station, including threshold ties.

20 of 53 evaluated stations have negative R-squared. Per-station results are included.

Previously inspected 2025 public benchmark; not a fresh independent test. No test targets enter training.

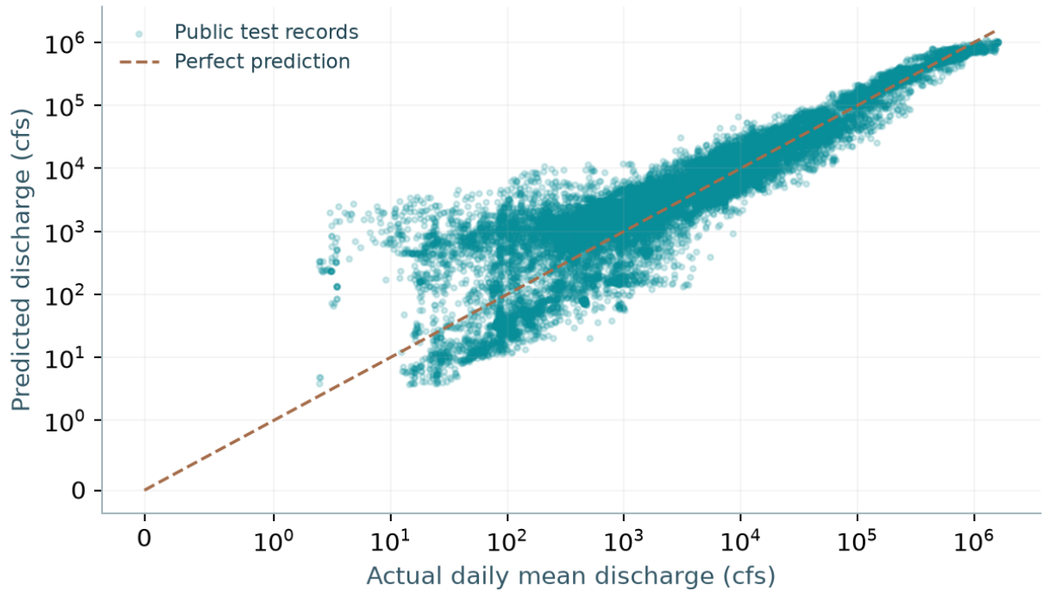
Sources: USGS discharge; Open-Meteo historical weather; USGS StreamStats and GAGES-II.

Full definitions, raw values, all predictions and limitations: evaluation/README.md and model_card.md

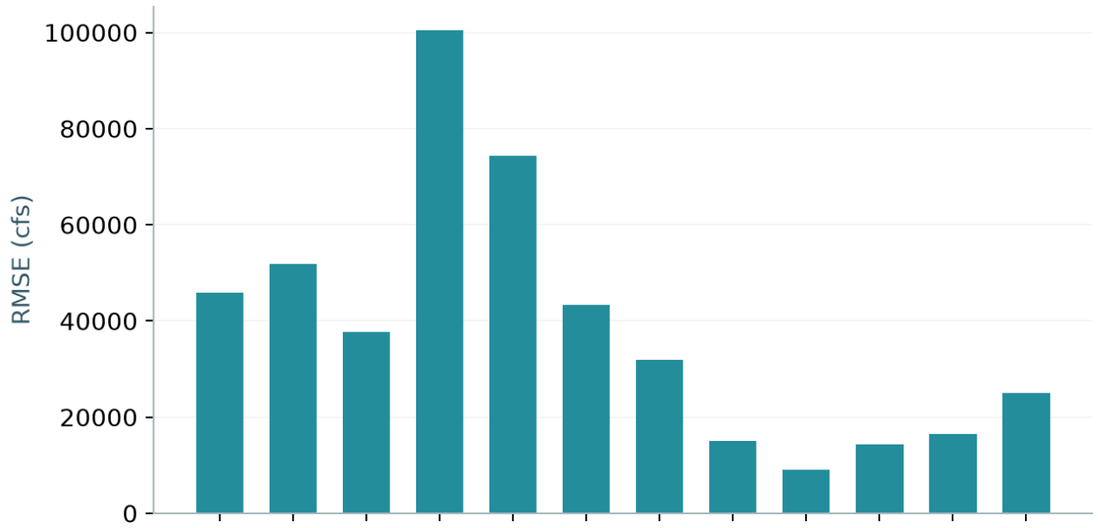
Actual predictions and test errors

FLOODCAST · frozen-model public test evaluation

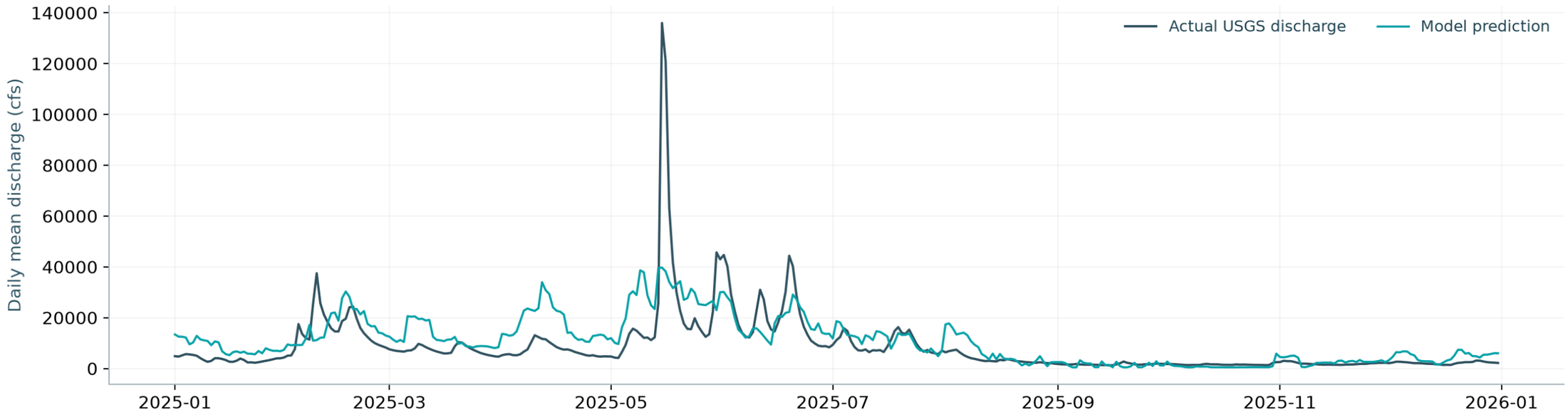
Actual vs predicted · all public test rows



Test error by month · 2025



Potomac station 01646500 · one named example, not the overall score



Every point comes from the frozen-model public test predictions. No invented curves or scores.